



COPILOTTRAINING
TECHNICAL BRIEFING

GitHub Copilot Code Review

| From First Feedback to a Human Decision

Get earlier PR feedback, then decide what needs human verification

35-40 min | Engineering Managers · DevOps Leads · Development Teams



"How can Copilot Code Review give a team useful feedback before the human review?"

A PR can collect actionable feedback while its author still has the change in mind.

Let Copilot propose findings; let people verify behavior, risk, and readiness.

Developer

Read early suggestions while the change is fresh



Engineering Manager

Compare review timing and rework in your own PR data



Security or Platform Team

Guide reviews with repository context; retain dedicated controls



PR

The review attaches to a specific diff and commit

Findings

Inspect the code and test a suggested fix

Decision

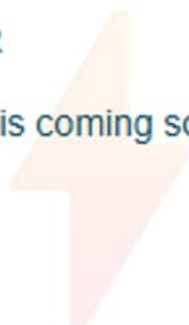
A human checks what Copilot missed before merging

PART 1

Configuration and Quick Start

Request a review on a real PR

Lite and Balanced today; Max is coming soon



PART 2

Best Practices and Adoption

Human questions, checks, and rollout

Equip teams to own the rollout and frame it for stakeholders

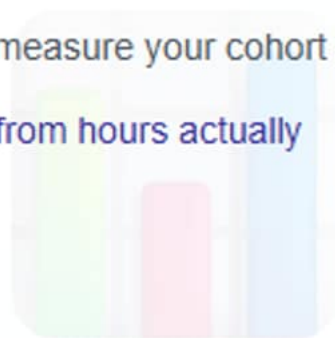


PART 3

Measuring Value

Model assumptions, then measure your cohort

Separate faster feedback from hours actually saved

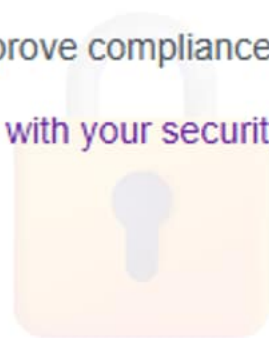


PART 4

Compliance Patterns

Instructions guide; controls prove compliance

Try scoped review guidance with your security owner



Configuration and Quick Start

Request one review, inspect its findings, then tune effort and instructions



First Review

Request from Reviewers or enable a ruleset



Effort Options

Choose what is available today



Custom Instructions

Encode team standards in Markdown

Start with one PR

↳ Request Copilot under Reviewers, or configure automatic review in a ruleset

Manual Review Workflow vs. Copilot-Automated Review

Early suggestions give the author and human reviewer a useful starting point

✗ Manual Review Workflow

- 1 **Submit PR**
Ask a teammate to review the change
- 2 **Read diff**
Trace behavior and repository context
- 3 **Discuss risks**
Test cases and boundary decisions emerge
- 4 **Human decision**
Accept or ask for changes



Human context and approval stay in the loop

✓ With Copilot Code Review

- 1 **Submit PR**
Request Copilot or enable automatic review
- 2 **Read findings**
Inspect inline suggestions against the code
- 3 **Test a correction**
Run a check that reproduces the concern
- 4 **Human review**
Verify the revised diff and remaining risks



Copilot gives an earlier, inspectable signal

Earlier feedback → tested correction → human decision

Three Effort Levels: Lite, Balanced, Max



Lite · available

Fast, targeted feedback on common issues.

Use: routine PRs where speed matters

AI-credit estimate: \$0.05–\$1 per review



Balanced · available

Longer analysis of complex logic and cross-service changes.

Use: security-sensitive or multi-service PRs

AI-credit estimate: \$0.25–\$5 per review



Max · coming soon

Most thorough level; not selectable yet.

Potential use: high-stakes PRs once available

No published credit estimate

⚙️ GitHub AI-credit estimates vary by PR; Actions minutes are extra. Default (Balanced) is not a fourth level.

Custom Instructions: Encode Your Review Standards

`.github/copilot-instructions.md`

Security Standards

- Flag hardcoded secrets and API keys
- Require parameterized queries (no SQL concatenation)
- Check input validation on user-facing code

Code Quality

- Suggest refactoring for functions exceeding 50 lines
- Flag unclear variable names

Testing

- Note missing unit tests for new functions
- Flag assertions that do not validate the logic

Prioritize top rules

Keep instructions concise; verify a review actually follows them

Language-specific files

`.github/instructions/*.instructions.md` with `applyTo` patterns

Layer guidance

Organization guidance and repository files can express different standards

Four Choices for Using Copilot Code Review



Repository Ruleset

Configure automatic review; choose draft and new-push triggers explicitly



Organization Default

Set review effort centrally; repositories can override the inherited level



Merge Policy

Normally COMMENTED; preview approvals can count if enabled. Keep human approval in this pilot



Manual Request

In the PR Reviewers menu, select Copilot → Request; re-request after a fix if needed

Best Practices and Team Adoption

Give the review context, pair it with other checks, and learn from a small pilot



Review Questions

Security to architecture consistency



Complementary Checks

Tests and scanning reveal different gaps



Phased Rollout

Pilot, measure, then decide where to expand

Copilot provides review suggestions

↳ Tests, scanning, and human judgment supply independent evidence

Six Questions for the Human Reviewer

Copilot may suggest findings; it does not certify that it answered every question

Security	Did the change introduce an unsafe input or secret?	ASK
Code Quality	Is a new pattern hard to maintain?	ASK
Tests	Would a focused test catch the suspected behavior?	PROVE
Performance	Does the changed path add expensive work?	MEASURE
Policy	Does this need a dedicated compliance or security check?	ESCALATE
Architecture	Does the change follow the repository contract?	COMPARE

Copilot excludes many lockfiles, config and generated files. Inspect those separately.

Three Signals, Different Evidence



Build and Tests

FanHub #198: four reported tests passed, but Retry was inert

A new render test failed on the first head

6/6 focused tests passed after correction

Frontend CI built; it did not run those tests



Code Quality

A separate PR scan raised five first-head findings

One broad catch in Home

Four test-client disposal patterns

New-head scan completed; inspect findings



Copilot Review

Automatic review found the browser click and alert problems

Static page: Retry could not respond

Alert vanished when retry began

Human tested the fix; re-request review

Phased Rollout: Pilot to Organization

Four phases from first review to org-wide standard

First pilot

Try

One or two repos —
compare suggestions with
actual code and tests



After feedback

Tune

Prioritize instructions; note
useful findings and missed
bugs



When useful

Expand

Train reviewers on triggers,
costs, and finding
disposition



With evidence

TARGET

Standardize

Revisit policy and review
timing with repository
owners



Suggested sequence, not a measured rollout schedule; keep merge policy separate from advisory comments

PART 3

Measuring Value in Your Team

Use the calculator for a scenario; use your own cohort to learn what changed



Scenario Calculator

Editable inputs and assumptions



Cycle Time

Compare comparable PR cohorts



Quality Signals

Incidents, reverts, reviewer effort

19 days vs. 4 days is a model input

↳ Calculate the difference; do not attribute it to Copilot without a study

Model the Opportunity; Verify the Inputs

Example inputs: 19 days vs. 4 days; results depend on the cohort and assumptions

78.9%

calculated difference between example cycle-time inputs

Illustrative calculator scenario — not a measured Copilot effect

Choose a cohort

Same repo, similar PR size, comparable calendar period

Show assumptions

Example: 100 PRs, 45 → 25 review minutes, \$100/hour

Read the output

Separate observed timing from modeled labor savings

Validate with owners

Ask which other process changes affected cycle time

 Days open and minutes of human effort are different measures. Only the second supports modeled labor savings.

What Would a Faster Review Cycle Mean?

Example cohort A

19 days open

Illustrative baseline, not a published benchmark

Measure time to first review

Count handoffs and revision rounds

Record minutes of human review effort

Example cohort B

4 days open

Illustrative comparison, not a Copilot effect

Check PR size and reviewer availability

Record useful suggestions and rework

Verify security with dedicated checks

15 days

example difference

78.9%

calculated change

?

causal impact to establish

Which Quality Signals Would You Track?



Production Incidents

Tag incidents linked to reviewed changes; compare comparable periods and severity.



Revert Rate

Count merged PRs later reverted, alongside review findings and test gaps.



Security Findings

Track verified issues from dedicated security tools and human review.



Time to Feedback

Measure first review, revisions, human effort, and time to merge separately.

From Scenario to a Question Worth Measuring

Illustrative calculator output — validate every input before reporting savings

```
$ copilot-code-review-calculator
> "Example: 100 PRs/month; 45 → 25 review min; $100/hour"
📊 Calculator:
100 × (45 - 25) / 60 = 33.3 review hours
33.3 × $100 ≈ $3,333 modeled labor value
19 → 4 days open is a separate cycle-time example
```

- ✓ Scenario output: ~33 hours, ~\$3,333; neither observed
- ✓ Next: compare matched cohorts and validate effort with reviewers

Scenario first → measured evidence before any ROI claim

Compliance and Regulatory Guidance

Use review guidance to ask better questions; let the control owner verify compliance



Health Data

Check the approved data boundary



Payment Data

Inspect handling and test evidence



Audit Controls

Escalate exceptions to the owner

Instructions focus an advisory review

↳ Compliance needs verified controls, evidence, and an authorized decision



Three Contexts, One Review Habit



Health Data

Ask where sensitive data flows and who approved the handling pattern

Inspect the documented boundary

Test denied access and redacted logs

Ask the privacy owner to validate



Payment Data

Ask whether the change follows the approved gateway integration

Inspect token and webhook handling

Run payment-path tests

Escalate exceptions to the control owner



Audit Controls

Ask what evidence the organization requires for access changes

Check logging and retention policy

Inspect actual audit events

Record human control approval

Compliance Instruction File Pattern

```
.github/instructions/health-data.instructions.md
```

```
---
```

```
applyTo: "dotnet/Frontend/**/*.razor"
```

```
---
```

```
# Review guidance (illustrative)
```

- For touched health-data paths, identify the documented access and logging controls.
- Ask for a test of denied access or redaction.
- Flag an unexplained departure from the repository pattern for a human reviewer.
- Escalate policy interpretation to the privacy and security owners.

applyTo pattern

Target the changed paths where this guidance is relevant.

Inspect output

Copilot may raise a useful question; run the actual controls and tests.

Human evidence

Record test results and the control owner's decision under your audit policy.

From One Review Signal to a Reviewable Decision

Before

A PR needs an available reviewer

The author knows the changed path

Checks cover named behaviors

The team has repository conventions

After

Copilot offers an early, advisory review

The author tests useful suggestions

Dedicated scanning checks other risks

A human validates the change and policy

1 PR

pilot the review on a consequential change

3 signals

review, tests, and scanning have different scopes

1 owner

a human decides whether the evidence is enough

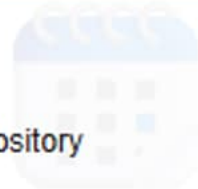
✓ WHAT YOU CAN DO TODAY

Today



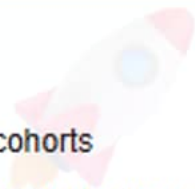
- Request Copilot from the PR Reviewers menu
- Use Default (Balanced), or choose Lite for routine changes
- Inspect one finding against the actual code

This Week



- Add one concise repository instruction
- Enable draft or new-push review deliberately
- Try Fix with Copilot if enabled; test and re-request review

This Month



- Compare similar PR cohorts
- Ask reviewers about useful findings and missed bugs
- Expand only when the evidence and owners support it

🔑 Key Takeaway

Start with one PR. Ask what changed, what was tested, and who accepts the remaining risk.

 Official Documentation[About Copilot code review](#)

Effort use cases, estimated AI-credit ranges, and limitations

[Configure reviews and effort](#)

Three levels; Default is a menu choice and Max is coming soon

[Request and interpret a review](#)

Reviewer menu, COMMENTED state, and re-review

[Write review instructions](#)

Focused guidance, path-specific files, and limitations

[Review file exclusions](#)

Lockfiles, configs, generated output, and other skipped files

 Interactive Tools[Time-savings scenario calculator](#)

Sign-in may be required; outputs depend on editable assumptions

 Announcements[Lite and Balanced availability](#)

GA names, inheritance, and per-review choice

[Balanced default and review API](#)

Current default and supported programmatic request

 Worked Example[FanHub draft #198](#)

Four tests passed; Copilot caught inert browser Retry



GitHub Copilot Code Review

FROM FIRST FEEDBACK TO A HUMAN DECISION

1 PR

request and inspect a review

1 finding

reproduce and test a correction

1 decision

a human accepts or asks for more evidence

What would you test before accepting Copilot's next useful finding?